

Control Valve Market Set to Surpass \$17.6 Billion by 2035

The global [Control Valve Market](#) was valued at USD 10.9 billion in 2025 and is projected to reach USD 17.6 billion by the end of 2035, expanding at a compound annual growth rate (CAGR) of 5.5% during the forecast period (2026–2035).

The market continues to experience stable expansion due to increasing industrial automation, modernization of process industries, and growing investments in energy, water treatment, chemicals, pharmaceuticals, and food processing facilities. Rising adoption of smart manufacturing technologies, digital process control systems, and Industrial Internet of Things (IIoT)-enabled equipment is further accelerating the demand for advanced control valves across both developed and emerging economies. Continuous replacement of aging industrial infrastructure and the increasing focus on operational safety, energy efficiency, and emission reduction are expected to sustain long-term market growth.

Control Valve Industry Demand

The control valve market represents one of the most critical segments of industrial process automation. Control valves regulate the flow rate, pressure, temperature, and liquid level within pipelines by automatically adjusting valve positions according to signals received from controllers. They are key elements in automated industrial systems, ensuring precise fluid control to enhance product quality, improve operational performance, and maintain workplace safety.

Control valves play a vital role in a wide range of industries, including oil and gas, chemical manufacturing, water and wastewater treatment, pharmaceuticals, power generation, mining, food and beverage processing, pulp and paper, and marine operations. As manufacturing facilities increasingly adopt automated production environments, the need for highly responsive, durable, and intelligent control valves continues to expand.

Growing emphasis on industrial digitalization has significantly increased the deployment of smart control valves equipped with sensors, digital positioners, predictive diagnostics, and remote monitoring capabilities. These innovative technologies enable industries to reduce equipment downtime, streamline maintenance planning, enhance energy efficiency, and lower overall operating expenses..

Demand is also supported by the expansion of renewable energy infrastructure, stricter industrial safety regulations, increasing investments in municipal water treatment projects, and the modernization of aging industrial plants. Industries are actively replacing conventional manually operated valves with automated control systems capable of delivering higher process precision and reliability.

Additionally, control valves provide several operational benefits including improved process efficiency, reduced labor requirements, enhanced plant safety, lower maintenance expenses, precise process regulation, longer service life, and compatibility with digital control platforms. Their ability to improve production consistency while minimizing energy losses has made them indispensable across modern industrial facilities.

Control Valve Market: Growth Drivers & Key Restraint

Growth Drivers –

Rapid Expansion of Industrial Automation- Industrial automation remains the strongest growth catalyst for the control valve market. Manufacturing industries are increasingly integrating programmable logic controllers (PLCs), distributed control systems (DCS), supervisory control and data acquisition (SCADA), and Industrial Internet of Things (IIoT) technologies into production facilities. Control valves act as essential field devices within these automation ecosystems by delivering accurate fluid regulation. Growing adoption of smart factories and Industry 4.0 technologies continues to create significant demand for digitally connected control valves capable of supporting predictive maintenance and real-time operational monitoring.

Increasing Investments in Energy and Process Industries- Rising investments in energy and process industries are significantly boosting market growth. Increased spending on oil and gas exploration, petrochemical facilities, power generation plants, hydrogen projects, water treatment infrastructure, and chemical manufacturing is driving the demand for advanced industrial solutions. Every large-scale industrial processing facility requires reliable control valves for maintaining pressure stability, flow accuracy, and safe operating conditions. Rapid industrialization across developing economies further strengthens long-term demand for advanced valve technologies.

Rising Demand for Smart and Energy-Efficient Process Equipment- Industries worldwide are prioritizing energy conservation, operational efficiency, and emission reduction. Smart control valves equipped with digital positioners, intelligent diagnostics, wireless communication, and predictive analytics enable better process optimization while minimizing energy consumption and maintenance costs. Growing implementation of asset management software and digital plant monitoring systems continues to increase the adoption of intelligent valve solutions.

Restraint –

High Initial Installation and Maintenance Costs

Despite strong market potential, the relatively high capital investment associated with advanced automated control valve systems remains a key market restraint. Intelligent control valves require sophisticated actuators, sensors, communication modules, and integration with automation software, increasing overall installation costs. In addition, regular calibration, specialized maintenance, and skilled technical personnel are required to ensure optimal system performance, limiting adoption among small and medium-sized manufacturing facilities with constrained capital budgets.

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Control Valve Market: Segment Analysis

Segment Analysis by Motion Type –

Linear Valves

Linear valves constitute a significant portion of the control valve market owing to their superior flow regulation accuracy and excellent sealing performance. These valves operate through linear movement of the valve stem, making them highly suitable for applications requiring precise flow modulation. Industries such as power generation, chemical processing, pharmaceuticals, and water treatment

extensively utilize linear valves because of their ability to maintain stable process conditions under varying operating environments. Continuous improvements in actuator technology and digital control systems further enhance their market adoption.

Rotary Valves

Rotary valves have gained widespread acceptance due to their compact design, rapid response time, and lower maintenance requirements. These valves regulate flow through rotational movement and are commonly employed in applications demanding high flow capacity with quick operational response. Oil & gas processing, food manufacturing, mining, and wastewater treatment industries increasingly prefer rotary valves because they offer excellent durability, simplified installation, and reliable performance under harsh industrial conditions. Their compatibility with automated systems continues to support growing demand.

Segment Analysis by Component –

Actuators

Actuators represent one of the most technologically advanced components within modern control valve systems. They convert control signals into mechanical movement that regulates valve operation. The growing transition toward electric and smart actuators has significantly improved process accuracy, remote operation, and predictive maintenance capabilities. Increasing adoption of automated industrial facilities continues to strengthen demand for advanced actuator technologies.

Valve Body

The valve body serves as the primary structural component responsible for containing process fluids under varying pressure and temperature conditions. Continuous innovations in corrosion-resistant materials, high-performance alloys, and advanced manufacturing techniques have improved durability, reliability, and operational lifespan. Industries handling aggressive chemicals and high-pressure fluids continue to demand premium valve body designs capable of ensuring long-term operational safety.

Others

Other components, including valve positioners, controllers, sensors, seals, packing materials, stems, and communication interfaces, contribute significantly to overall system performance. The increasing integration of digital communication protocols and intelligent diagnostics has expanded the role of these auxiliary components, enabling improved monitoring, fault detection, and process optimization across industrial automation systems.

Segment Analysis by Application –

Flow Control

Flow control remains the largest application area for control valves, as accurate regulation of liquid, gas, and steam flow is essential across nearly every industrial process. Manufacturing facilities rely on efficient flow control to maintain production consistency, reduce waste, optimize resource utilization, and ensure product quality. Continuous industrial automation further strengthens demand within this application segment.

Pressure Control

Pressure control applications require highly reliable valve systems capable of maintaining stable operating conditions within pipelines and processing equipment. Industries such as oil & gas, power generation, petrochemicals, and chemical manufacturing depend on precise pressure regulation to ensure equipment safety, process stability, and regulatory compliance. Increasing emphasis on operational safety continues to support growth in this segment.

Other Process Controls

Other process control applications include temperature regulation, liquid level management, gas mixing, steam distribution, chemical dosing, and process isolation. Expanding deployment of integrated automation systems has increased the utilization of multifunctional control valves capable of supporting multiple process parameters simultaneously. These applications continue to gain importance as industries pursue higher operational efficiency and improved production flexibility.

Control Valve Market: Regional Insights

North America represents a technologically mature market supported by advanced industrial automation infrastructure and continuous modernization of manufacturing facilities. Strong demand originates from oil & gas production, power generation, pharmaceuticals, chemicals, food processing, and municipal water treatment sectors. The widespread adoption of Industry 4.0 technologies, smart manufacturing platforms, and predictive maintenance solutions continues to accelerate demand for intelligent control valves. Increasing investments in energy infrastructure modernization and environmental compliance further strengthen regional market expansion.

Europe maintains a strong market position due to stringent industrial safety regulations, environmental sustainability initiatives, and extensive deployment of automated manufacturing systems. Chemical processing, renewable energy, pharmaceuticals, food production, and wastewater treatment industries remain major consumers of advanced control valve technologies. Growing emphasis on reducing industrial emissions, improving energy efficiency, and supporting digital manufacturing transformation continues to encourage replacement of conventional valve systems with smart, digitally connected control valves. Continuous investments in hydrogen infrastructure and clean energy projects further contribute to market growth.

Asia-Pacific represents the fastest-growing regional market, supported by rapid industrialization, expanding manufacturing activities, and large-scale infrastructure development. Countries across the region continue to invest heavily in chemical production, power generation, water treatment, oil & gas processing, pharmaceuticals, mining, and food manufacturing facilities. Rising foreign direct investment, increasing factory automation, expanding smart manufacturing initiatives, and government support for industrial modernization significantly drive demand for advanced control valves. The ongoing expansion of industrial production capacity positions Asia-Pacific as the primary engine of future market growth.

Top Players in the Control Valve Market

The global control valve market is highly competitive, with leading manufacturers focusing on product innovation, digital valve technologies, automation integration, strategic acquisitions, and global distribution expansion. Major companies operating in the market include Emerson Electric Co. (U.S.),

Flowserve Corporation (U.S.), Baker Hughes Company (U.S.), Curtiss-Wright Corporation (U.S.), Crane Company (U.S.), CIRCOR International Inc. (U.S.), IMI plc (UK), Rotork plc (UK), Spirax-Sarco Engineering plc (UK), KSB SE & Co. KGaA (Germany), Christian Bürkert GmbH & Co. KG (Germany), SAMSON AG (Germany), Metso Corporation (Finland), Alfa Laval AB (Sweden), KITZ Corporation (Japan), and Trillium Flow Technologies (UK). These companies continue to strengthen their market positions through investments in smart valve solutions, advanced automation technologies, digital diagnostics, predictive maintenance capabilities, customized industrial solutions, and expansion into high-growth emerging markets.

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